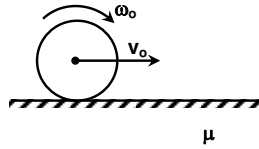
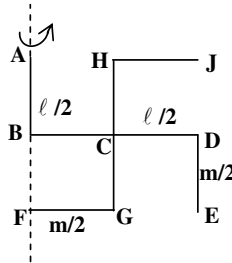


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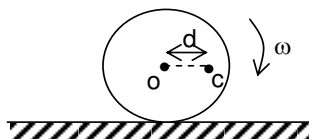
1. A sphere of mass M and radius R is moving on a rough fixed surface, having co-efficient of friction μ as shown in figure. It will attain a minimum linear velocity after at time



- (A) $V_0 / \mu g$ (B) $\omega_0 R / \mu g$
 (C) $(V_0 - \omega_0 R) / \mu g$ (D) $2(V_0 - \omega_0 R) / 7 \mu g$
2. A solid sphere mass 10 kg and radius 0.1 m is placed on a rough horizontal surface. A 20 N force is applied on the sphere horizontally passing through centre of mass. For pure rolling, the force of friction is
- (A) $\frac{40}{7}$ N (B) $\frac{20}{7}$ N
 (C) 20 N (D) none
3. As shown in the diagram a frame which is initially in the horizontal plane is rotated about 'AB'. Find the angular velocity of the system at the moment it comes in a vertical plane from the initial position. (Assume it being made up of uniform rod of length ℓ & $\ell/2$ (mass of length $\ell = m$)



- (A) $\omega = \sqrt{\frac{3g}{33\ell}}$ (B) $\omega = \sqrt{\frac{86g}{33\ell}}$
 (C) $\omega = \sqrt{\frac{48g}{17\ell}}$ (D) $\omega = \sqrt{\frac{8g}{7\ell}}$
4. A solid sphere is rolling on a rough surface, whose centre of mass is at C at a certain instant. Find at that instant it has angular velocity ω . Its radius is R . Find the angular acceleration at that instant mass of sphere is m



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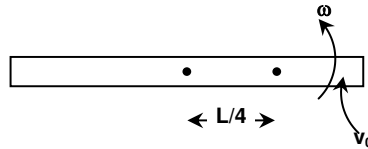
(A) $\frac{5}{2} \frac{gd}{R^2}$

(B) $\frac{5}{7} \frac{gd}{R^2}$

(C) $\frac{5}{2} \frac{d(g + \omega^2 R)}{R^2}$

(D) $\frac{5}{7} \frac{d(g + \omega^2 R)}{R^2}$

5. A stick of length ℓ and mass m lies on a frictionless horizontal surface on which it is free to move in any direction. A ball of same mass m moving with speed v_0 perpendicularly to this length and collide at the end of stick. Find the angular velocity when the collision is perfectly inelastic.



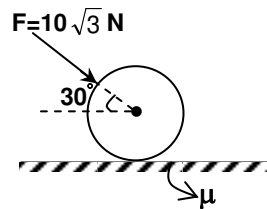
(A) $\frac{12v_0}{7\ell}$

(B) $\frac{3v_0}{2\ell}$

(C) $\frac{24v_0}{5\ell}$

(D) $\frac{3v_0}{7\ell}$

6. A hollow sphere of mass 2 kg is kept on a rough horizontal surface. A force of $10\sqrt{3}$ N is applied as shown in the figure. Find the minimum value of ' μ ' so that the sphere starts pure rolling. (Take $g = 10 \text{ m/s}^2$)



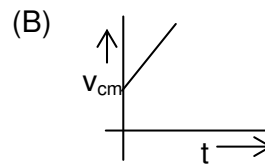
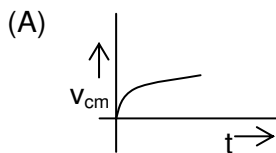
(A) 0.3

(B) 0.2

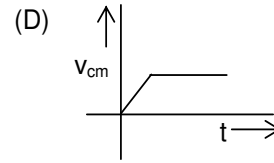
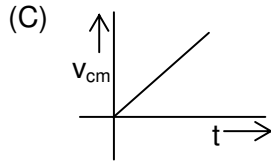
(C) 0.1

(D) for no value of μ it will start pure rolling

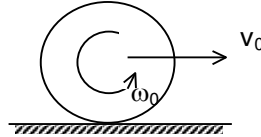
7. A uniform disc of radius r is rotated in clockwise sense with angular speed ω and kept vertically on a rough surface. Which of the graph of $v_{cm} - t$ is correct.



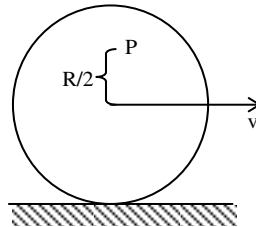
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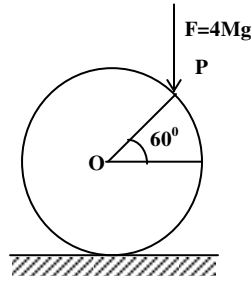
8. A uniform circular disc of radius r is placed on a rough horizontal surface and given a linear velocity v_0 and angular velocity ω_0 as shown. The disc comes to rest after moving some distance to the right. It follows that



- (A) $3 v_0 = 2 \omega_0 r$ (B) $2 v_0 = \omega_0 r$
 (C) $v_0 = \omega_0 r$ (D) $2 v_0 = 3 \omega_0 r$
12. A sphere cannot roll continuously on
 (A) a smooth horizontal surface. (B) a smooth inclined surface.
 (C) a rough horizontal surface. (D) a rough inclined surface.
13. A cylinder of mass ' M ' and radius r is rolling without slipping on a horizontal surface. The velocity of its C.M. is equal to v . Find the total angular momentum of the cylinder about point P on the cylinder at distance $R/2$ from centre



- (A) $\frac{MR^2\omega}{4}$ (B) $\frac{MR^2\omega}{2}$
 (C) $MR^2\omega$ (D) zero
14. A solid sphere of mass M and radius R is lying on a rough horizontal plane. A constant force $F = 4 Mg$ acts vertically at point P such that OP makes 60° with horizontal. Find the minimum value of coefficient of friction μ so that sphere starts pure rolling.



- (A) $\frac{3}{7}$ (B) $\frac{4}{7}$
 (C) $\frac{2}{7}$ (D) $\frac{2}{5}$

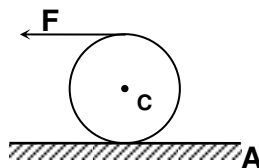
15. A rectangular block has a square base measuring $a \times a$, and its height is h . It moves on a horizontal surface in a direction perpendicular to one of the edges on the base. The coefficient of friction is μ . It will topple if

- (A) $\mu > h/a$ (B) $\mu > a/h$
 (C) $\mu > \frac{2a}{h}$ (D) $\mu > \frac{a}{2h}$

16. A uniform rod of length 2ℓ is placed with one end in contact with the horizontal table and is then inclined at an angle α to the horizontal and allowed to fall. Friction is sufficient so that the end in contact does not slip. When it becomes horizontal, its angular velocity will be

- (A) $\omega = \sqrt{\left(\frac{3g \sin \alpha}{2\ell}\right)}$ (B) $\omega = \sqrt{\left(\frac{2\ell}{3g \sin \alpha}\right)}$
 (C) $\omega = \sqrt{\left(\frac{g \sin \alpha}{\ell}\right)}$ (D) $\omega = \sqrt{\left(\frac{\ell}{g \sin \alpha}\right)}$

17. A massless string is wrapped around a hollow cylinder having mass m and radius r . The cylinder is kept on a rough horizontal surface (coefficient of friction is μ). A constant force F is applied as shown in the figure. In case of pure rolling, the friction force acting on the bottom most point of the cylinder is



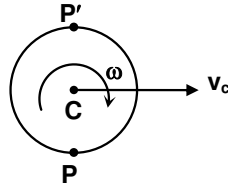
- (A) 0 (B) μmg

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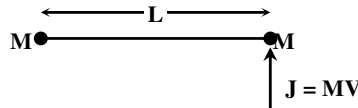
(C) $\frac{\mu mg}{2}$

(D) $\frac{3\mu mg}{2}$

18. A sphere is moving towards +ve x-axis with a velocity v_c and rotates clockwise with angular speed ω as shown in the figure, such that $v_c > \omega R$. The instantaneous axis of rotation will be



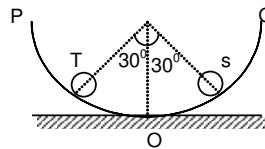
- (A) on point P
(B) on point P'
(C) inside the sphere
(D) outside the sphere
19. Consider a body, shown in figure, consisting of two identical balls, each of mass M connected by a light rigid rod. If an impulse $J = MV$ is imparted to the body at one of its ends, what would be its angular velocity?



- (A) V/L
(B) $2V/L$
(C) $V/3L$
(D) $V/4L$

SUBJECTIVE :

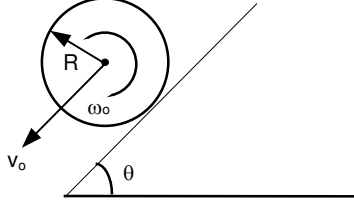
11. A small solid of mass ' M ' and radius ' r ' starts from rest at a point 'P' on the fixed hemispherical rough track of radius ' R ' and moves with pure rolling. The part 'PO' of the track has sufficient friction to maintain the rolling motion, whereas part 'OQ' of track is frictionless.



- (a) Find the normal force exerted on the sphere at point 'T' & 'S'.
(b) Find the maximum height reached in the portion 'OQ' of the track.

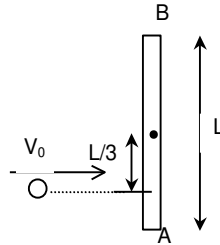
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12. A sphere of radius R is projected with a reverse spin ω_0 down a rough inclined plane with a speed v_0 for which coefficient of friction is $\mu > \tan \theta$ where θ is angle of the incline. Show that it will turn back if $\omega_0 > \frac{5v_0\mu}{2R(\mu - \tan \theta)}$

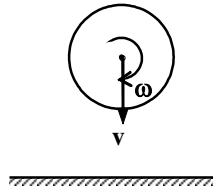


13. The linear mass density of the rod AB of length l is varying as $\lambda = \lambda_0 (1 + ax)$, where λ_0 and a are constants and x is the distance measured from end A. The rod is pivoted at the centre and is lying on a smooth horizontal surface. If an impulsive blow P strikes at B perpendicular to the rod then find the angular velocity attained by the rod.

14. A rod AB of mass M and length L is kept on a smooth horizontal floor as shown. A small object of mass ' m ' and velocity ' v_0 ' as shown moving perpendicular to the length of the rod hits the rod and as a result its velocity direction gets reversed with a magnitude of $v_0/2$ and velocity of centre of mass of the rod becomes $v_0/4$.

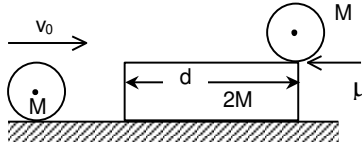


- (a) Find the ratio m/M
 (b) Find the location of a point P on the rod which acquires a velocity of $v_0/2$ immediately after collision.
15. A rotating ball hits a rough horizontal plane with a vertical velocity v and angular velocity ω . Given that the co-efficient of friction is μ and the vertical velocity of the ball after the collision is $v/2$, find the angular velocity after collision.



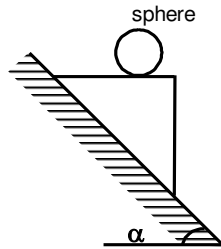
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16. A cylinder of mass 'M' is kept on top of a block of mass 2M. There is no friction between the block and the horizontal plane. But friction coefficient between the cylinder and the top surface of the block is μ .



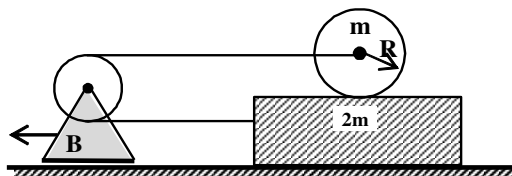
A small body of mass M travelling with a velocity v_0 hits the block horizontally. Collision is completely elastic. Find the minimum value of v_0 for which the cylinder does not start pure rolling before it topples from the top of the block.

17. A heavy sphere of mass $3m$ and radius R_0 is placed on a perfectly rough horizontal surface of a wedge of mass m as shown in figure. Determine the accelerations of the wedge and the sphere if the wedge can slide on a smooth plane inclined at an angle of $\alpha = 45^\circ$.

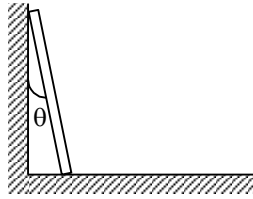


18. On a board of mass M lying in a state of rest on a smooth horizontal surface, a uniform solid sphere of mass m and radius R is projected with initial velocity v_0 & initial angular velocity ω_0 ($v_0 > R\omega_0$). Show that at the time $t = \frac{2M}{7M + 2m} \frac{(v_0 - R\omega_0)}{\mu g}$ the sphere starts rolling on the board and the velocity of the board becomes $\frac{2M}{7M + 2m} (v_0 - R\omega_0)$. Where μ is coefficient of friction between sphere and board.

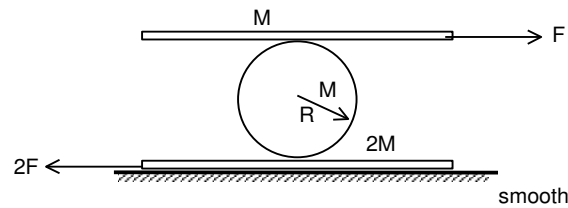
19. A solid cylinder of mass m and radius R rest on a plank of mass 2M lying on a smooth horizontal surface. String connecting cylinder to the plank is passing over a massless pulley mounted on a movable light block B and the friction between the cylinder and the plank is sufficient to prevent slipping. If the block B is pulled with a constant force F, find the acceleration of the cylinder and that of the plank.



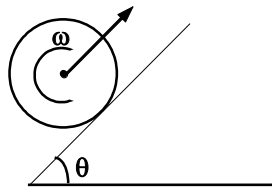
20. Find the value of θ when the rod breaks its contact with vertical wall. Initially $\theta \approx 0$. Ignore friction.



21. A cylinder is sandwiched between two planks. Two constant horizontal forces F and $2F$ are applied on the planks as shown. Masses of planks and cylinder are indicated in the figure. Radius of the cylinder is as shown. Find acceleration of the center of mass of cylinder and the top plank if there is no slipping at the top and bottom of cylinder.

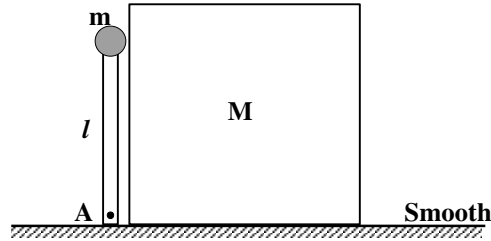


22. A sphere is projected up an inclined plane for which $\mu = (1/7) \tan \theta$ with a velocity v_0 and initial angular velocity ω_0 ($v_0 > R\omega_0$). Show that friction acts downward at first and upwards afterwards. Further prove that the total time of rise is $\frac{17v_0 + 4R\omega_0}{18g \sin \theta}$.



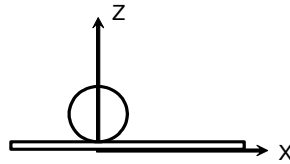
23. A massless rod of length l with a small load of mass m at the end is freely pivoted at point A. The rod is initially in a vertical position and is touching a block of mass M which rests on a smooth horizontal ground. The rod is given a slight jerk and it starts rotating about point A. This causes the block to move forward. After some time the rod forms an angle θ with the horizontal. at this instant the speed of the mass m is v and the magnitude of its tangential acceleration is a_t .

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- (a) Determine the velocity and acceleration of mass M . When $\theta = 30^\circ$, the normal force between m and M vanishes.
- (b) Find the velocity of mass M at this instant.
- (c) Find the value of (M/m) so that this happens.

24. A cylinder of mass M and radius R is kept on a rough horizontal platform at one extreme of the platform at $t = 0$. Platform is in x - y plane. Axis of the cylinder is parallel to Y -axis. The platform is oscillating in the x - y plane and its displacement from the origin is represented as $x = 2 \cos(4\pi t)$ m. There is no slipping between the cylinder and the platform. Find the acceleration of center of mass of the cylinder as a function of time.



25. A solid cylinder is kept on a rough horizontal surface and a plank is kept on it. A horizontal force F is applied on the plank. Mass of the plank and the cylinder is M each. If radius of the cylinder is R . Find the acceleration of the centre of mass of cylinder and the plank, if there is no slipping at any contact.

